

Critical Design Review (CDR): TerraProbe

A Down2Earth (D2E) Company

3/11/2025



Meet the Team



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PDR Recap – Problem & Customer Requirements

Soil Testing Challenges: Costly, Slow, and Inaccurate

- **Soil Testing in Agricultural Industry**
 - Important!
 - Soil testing process is not optimal
 - Inaccurate, timely, expensive
- **Customer Requirements (Farmers)**
 - Easy to operate
 - Accuracy in data
 - Portable
 - Time Saving



PDR Recap - Competitive Landscape

Current Soil Sampling Solutions are expensive, complex, and outdated

		Portability	Cost	Sampling Depth	Operation Type	Durability	Real-Time Data
	AMS Soil Probes	Weight: ~1 to 3 lbs Length: ~33 inches	\$100-\$150	24"	Manual	Stainless Steel, requires minimal maintenance	No
	WintexAgro 1000 Automatic Soil Sampler	Weight: 105 lbs, Requires mount on Vehicle	\$8000	11.8"	Designed for vehicle sampling, automated sampling	Requires heavy maintenance due to bulky equipment	No
	Amity Technology Soil Sampler	Weight: 210 lbs Requires mount on vehicle	\$3000-\$6000	24 to 48" (Based on model)	Designed for vehicle mounting, automated sampling	Requires standard maintenance	Yes, partners with 3 rd party to provide (FARMQA)

Market Gap for an Intermediate Solution

- **Cost:** High-end solutions cost \$5,000+, while cheaper alternatives lack precision and automation.
- **Operation:** Advanced models require skilled operation, while manual tools are slow and labor-intensive.
- **Insights:** Lab-based testing takes days, preventing real-time decision-making for farmers and researchers.

PDR Recap – Design Selection Process

Design process emphasized balancing soil depth profiling/testing, portability, and ease of use

Concept 1

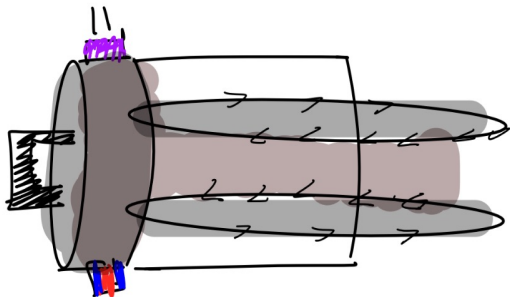
Function:

- Pressurized Gas, evertng system
- Can get jammed

Decision Matrix:

- Sampling Accuracy: 5
- Durability: -4
- Precision in depth collection: 5

Total: 36



Concept 2

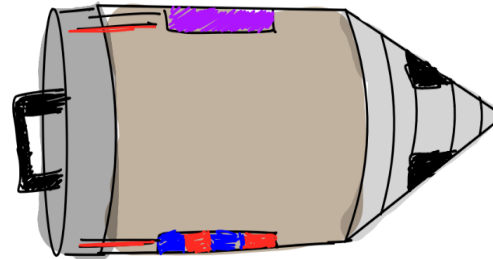
Function:

- Drill and collect
- High Torque, Can get lodged

Decision Matrix:

- Sampling Accuracy: -5
- Durability: 0
- Precision in depth collection: 5

Total: 23



Concept 3 (Selected)

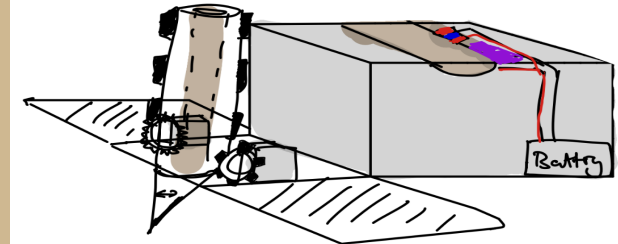
Function:

- Pushes in ground, collects sample
- Separate Data Acquisition system

Decision Matrix:

- Sampling Accuracy: 10
- Durability: 4
- Precision in depth collection: 10

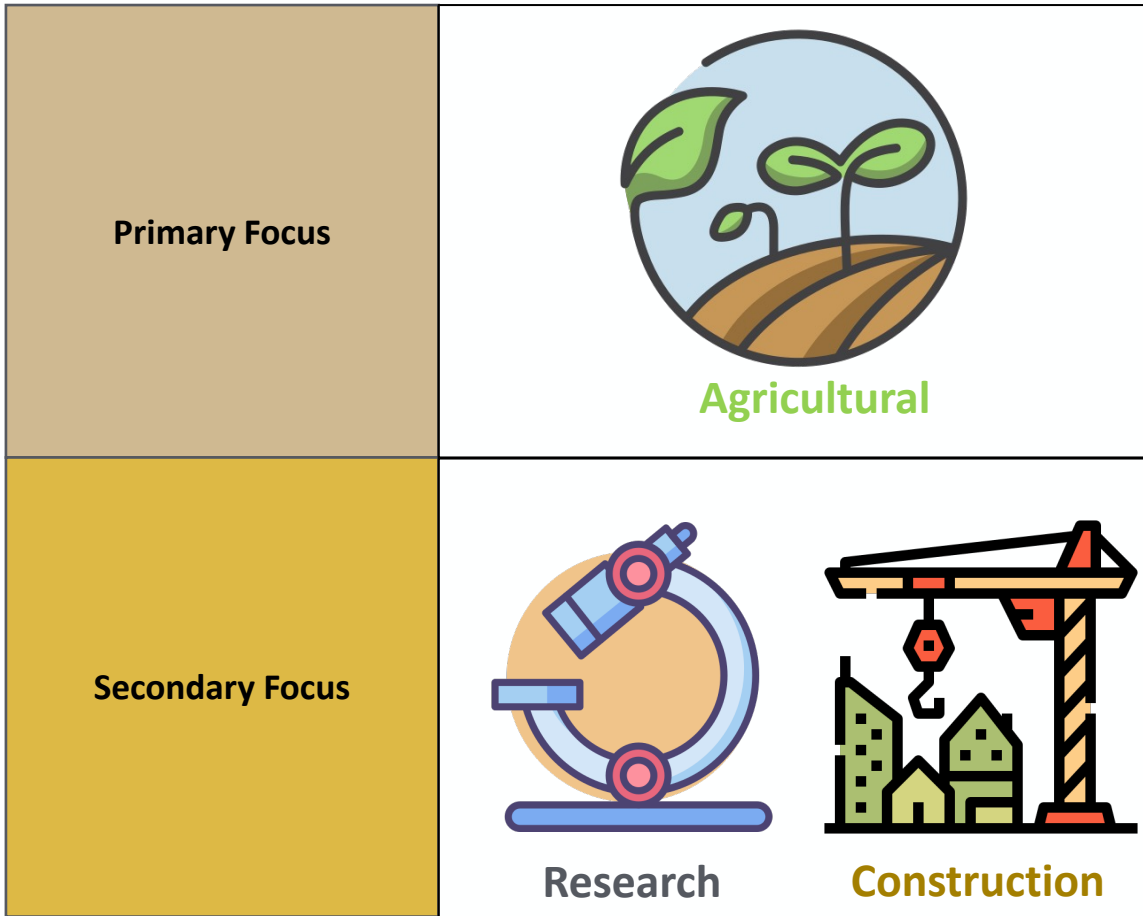
Total: 52



Major Changes Since PDR

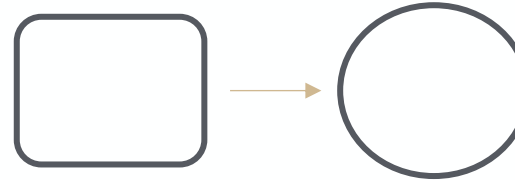
Focused target customers and made appropriate mechanical and software changes

Streamlined Target Market

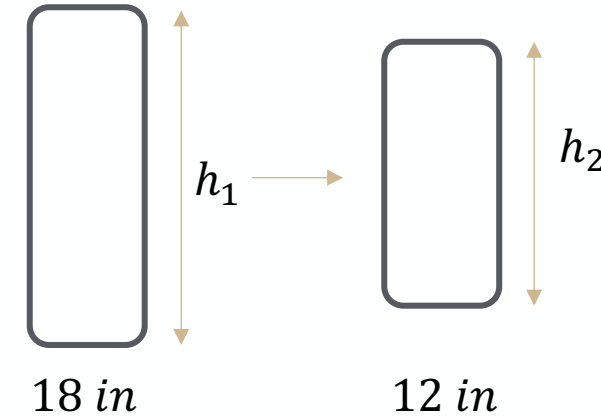


Mechanical Changes

Payload cross-section:



Payload Height :

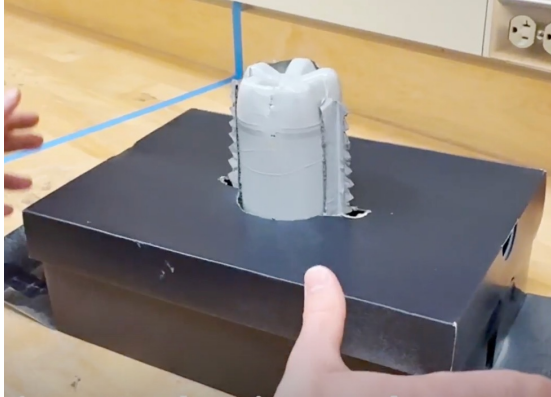


Software Changes

- Removed Temperature Sensor
- Changed from ESP 32 to Arduino UNO
- Wireless Dashboard to see live sensor results and preliminary analysis

PDR Recap – Prototype & Learnings

Focused target customers and made appropriate mechanical and software changes



TerraProbe Assembly



Motor-Shaft-Gear System



Soil Testing Probe



Testing & Slits Design

Key Learnings



Converted Foot
Pedals into Dual
Purpose Handle



Added Bearings for
Increased Safety



Determined Spacing
& Wire Management
for Electronics



Increased Slit Height
& Width for Sensor
Accommodation

Key Product Features & Functions

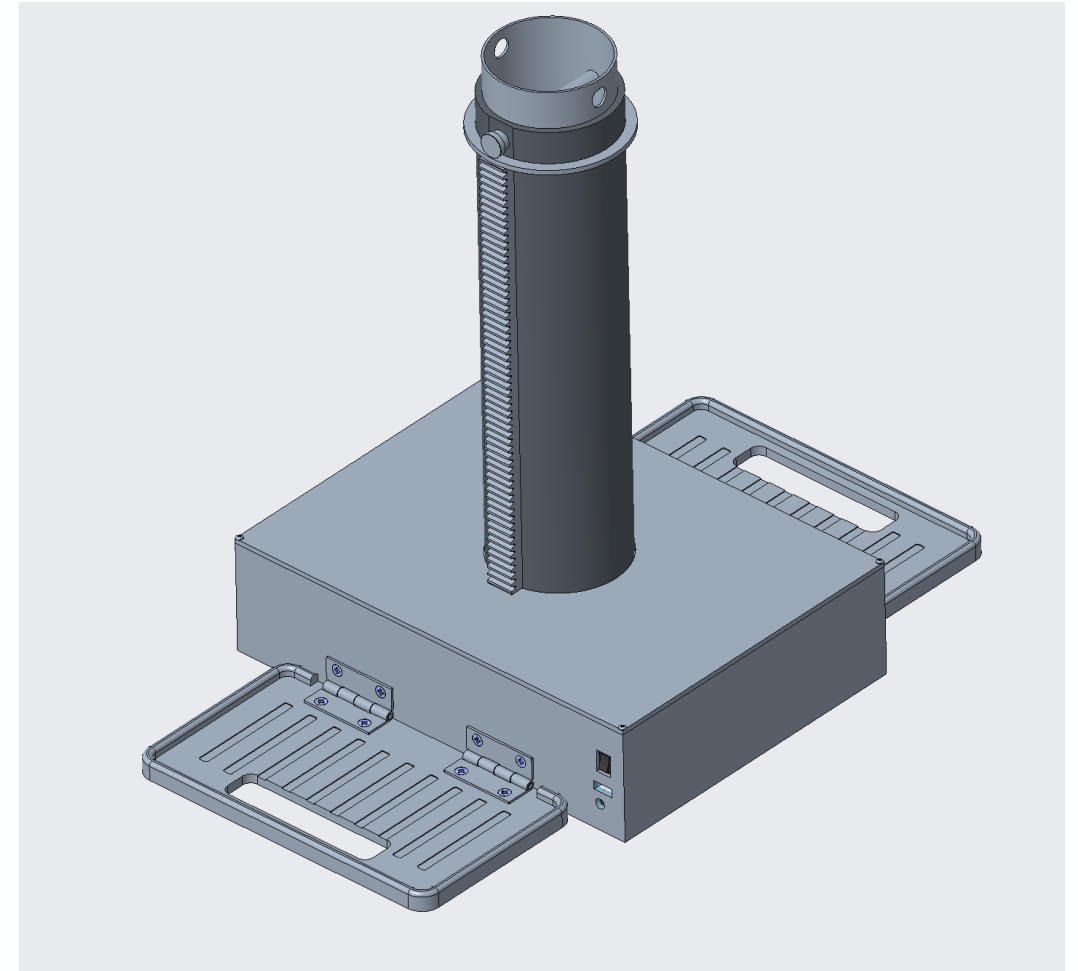
TerraProbe presents a compact, affordable, user-friendly product, with real-time data insights.

Customer Design Features:

- Compact Design: 10" x 12" x 17", 35 lbs. for easy portability
- Motor-Controlled System: Rack and pinion mechanism drives payload into soil
- User-Friendly Features: Foldable foot pedals for added force, removable top lid for maintenance
- Integrated Testing Probe: Measures moisture, NPK levels for real-time data analysis
- Modular Inner Payload: Replaceable soil collection chamber

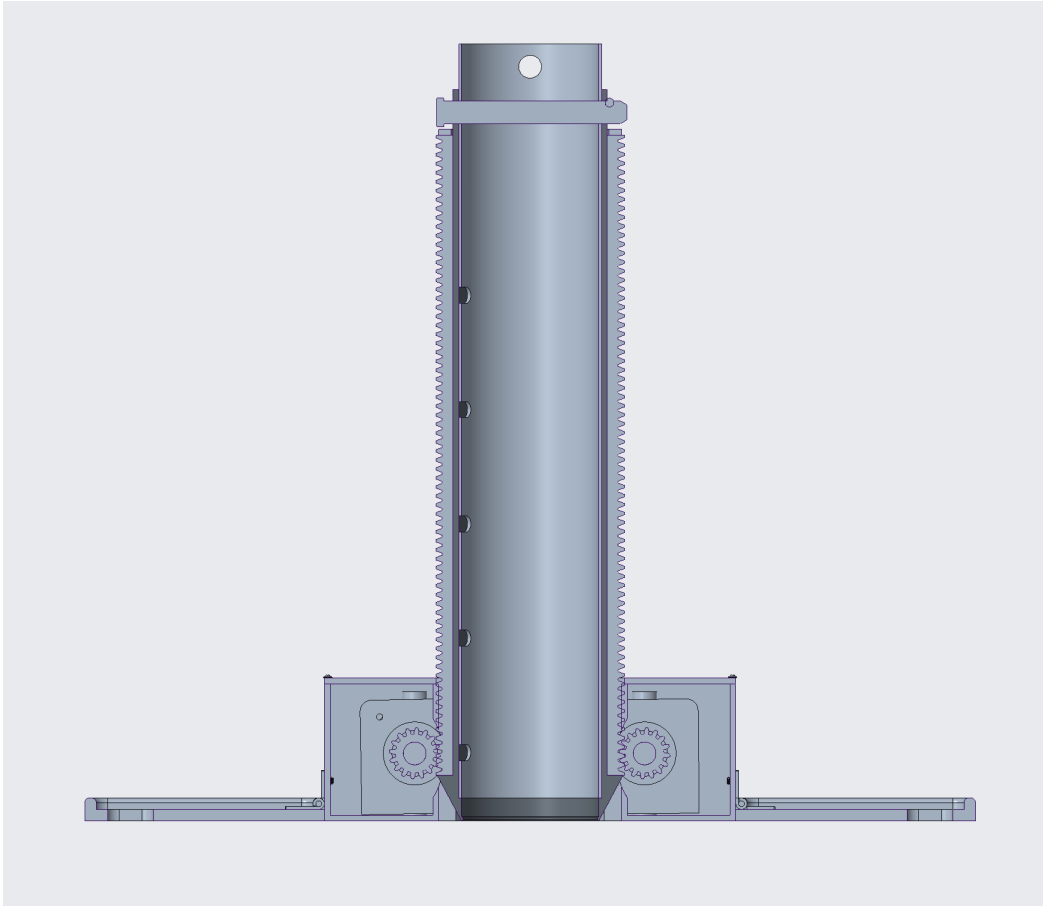
Mechanical Design Features:

- Rack and pinion: Drives payload and outer shell into the ground
- Foldable foot pedals: Allow users to apply downward force for improved penetration.
- Removable top lid: Provides easy access for maintenance and cleaning.
- Limit switch mechanism: Prevents over-extension of the outer shell during operation.

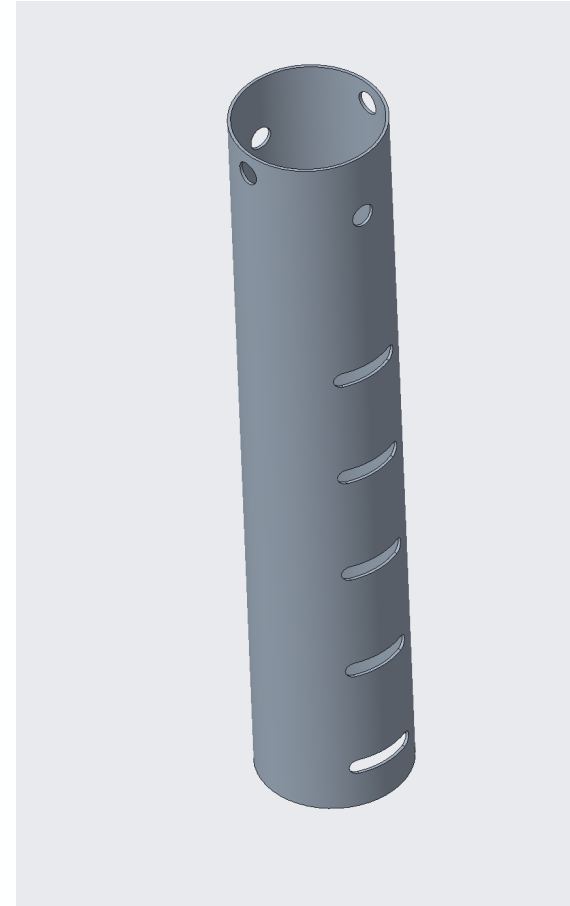


TerraProbe's Key CAD Components

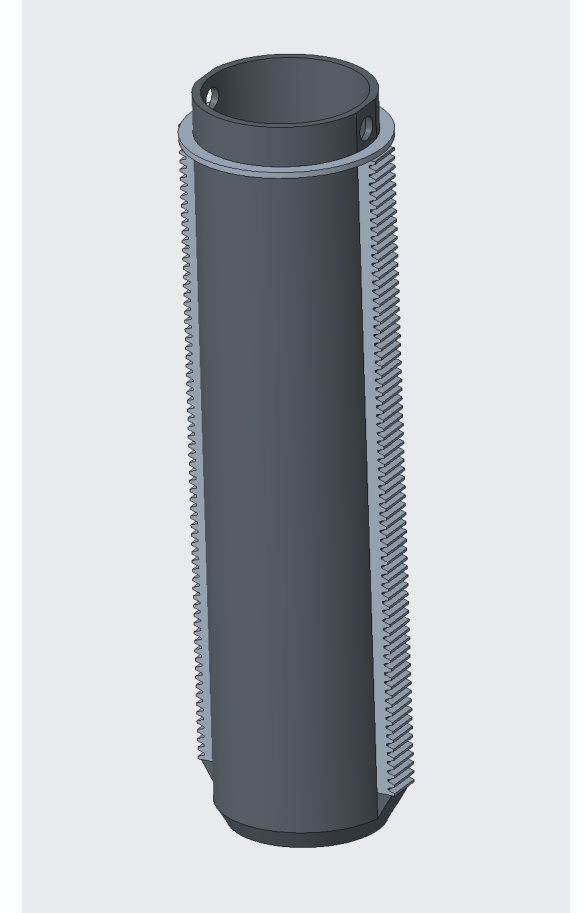
Unique Payload and Rack/Pinion Design for Soil Sampling



Base



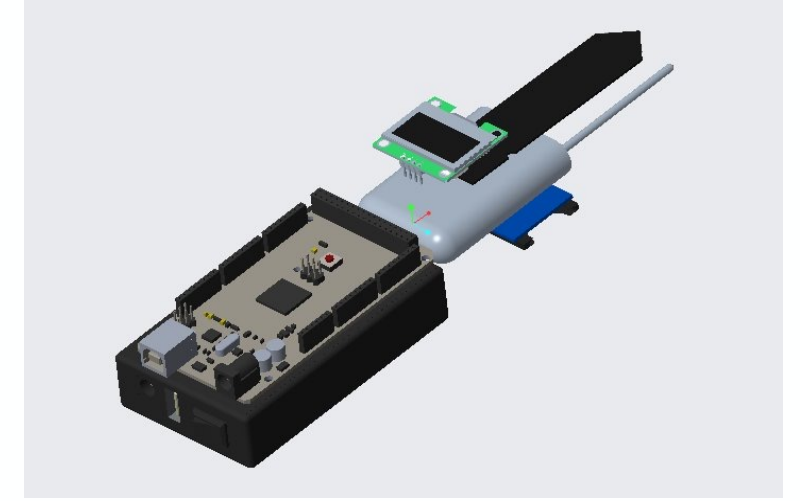
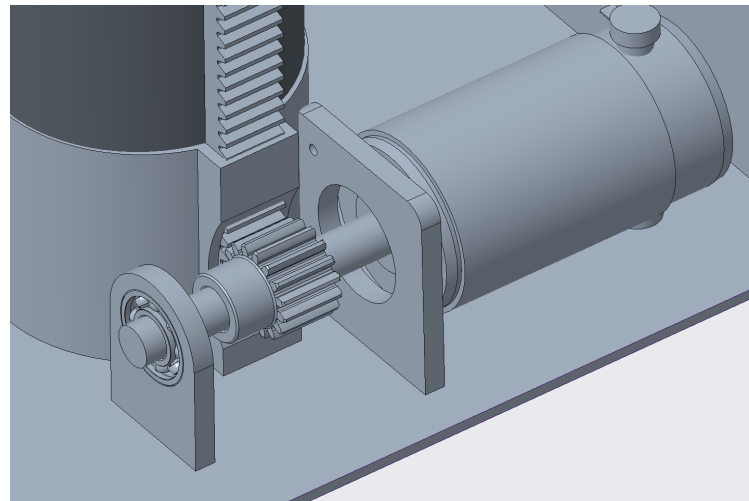
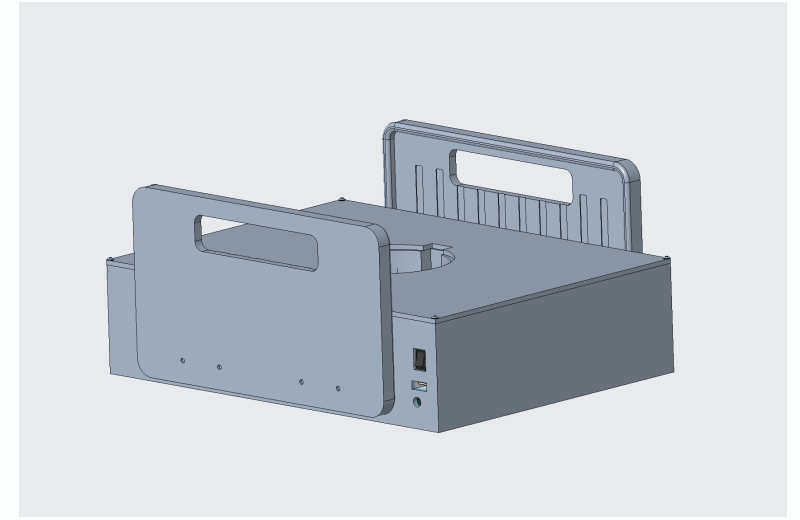
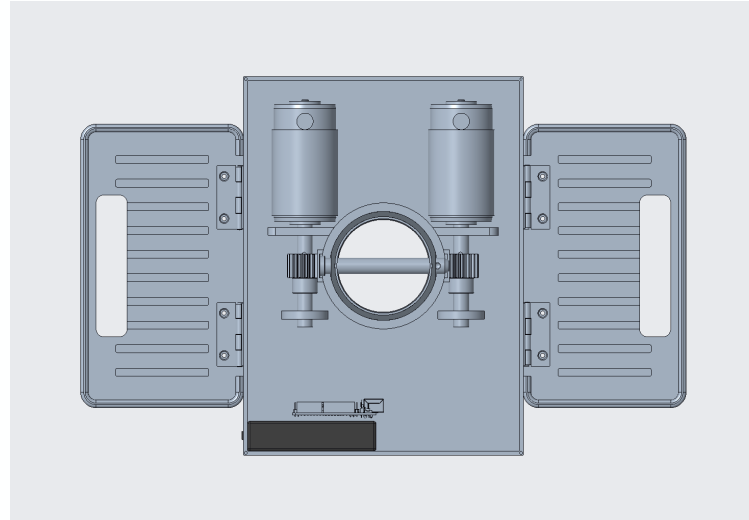
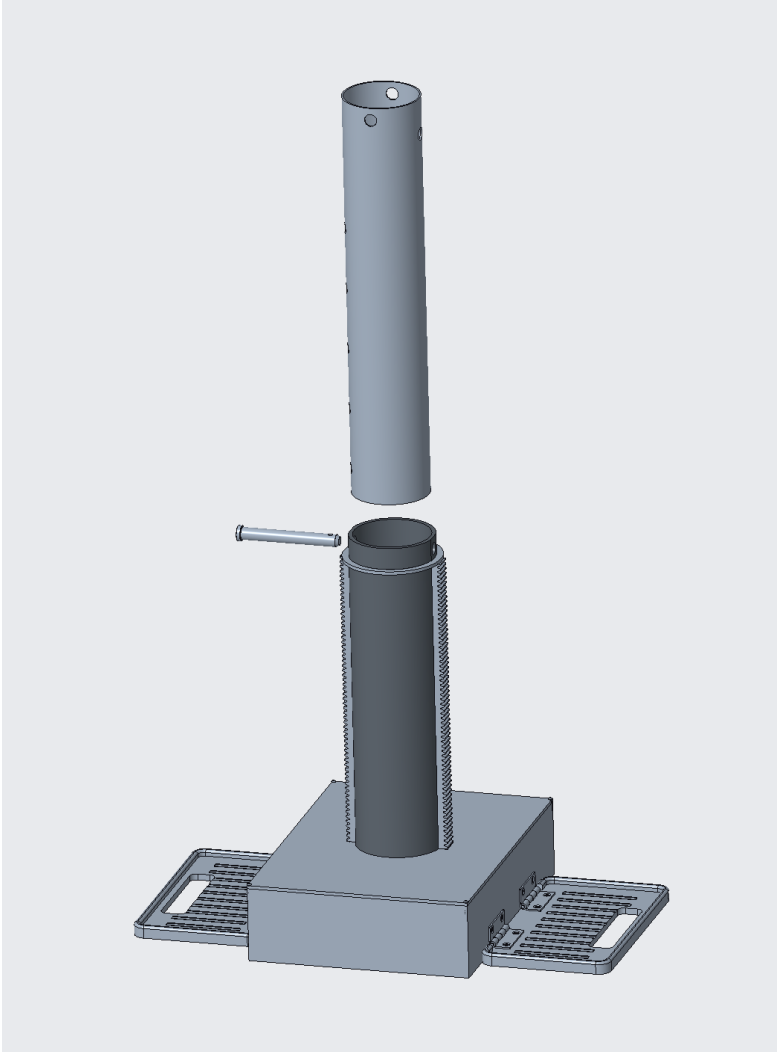
Payload



Shell and Racks

TerraProbe's Sub-Assemblies

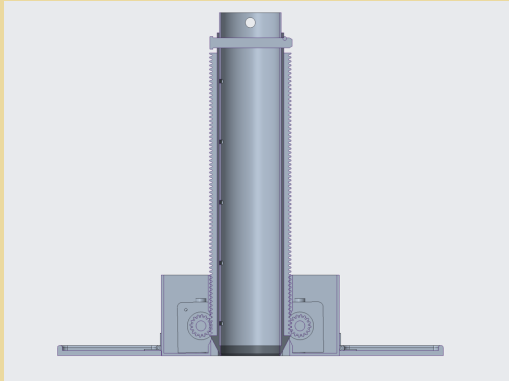
Payload for sampling, gear-shaft for power, and probe for real-time analysis.



Pricing & Competitive Advantage

The model was adjusted to the new requirements and the required force was determined

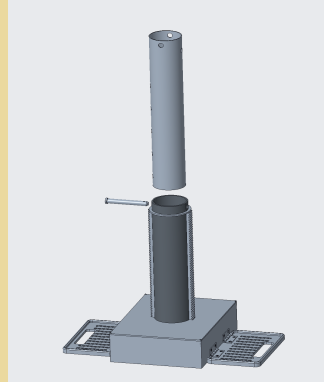
TerraProbe DEVICE



Portable & Automated
Soil Burrowing System &
Testing Probe

Sell Price: \$1050 per unit

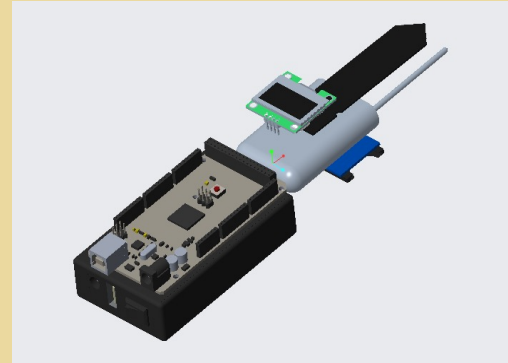
MODULAR INNER TUBES



Replicable inner component
for multi-depth **sampling**
and **collection**

Sell Price: \$130 per unit

ANALYTICS SUBSCRIPTION



Real-time soil health insights
via a **digital dashboard**

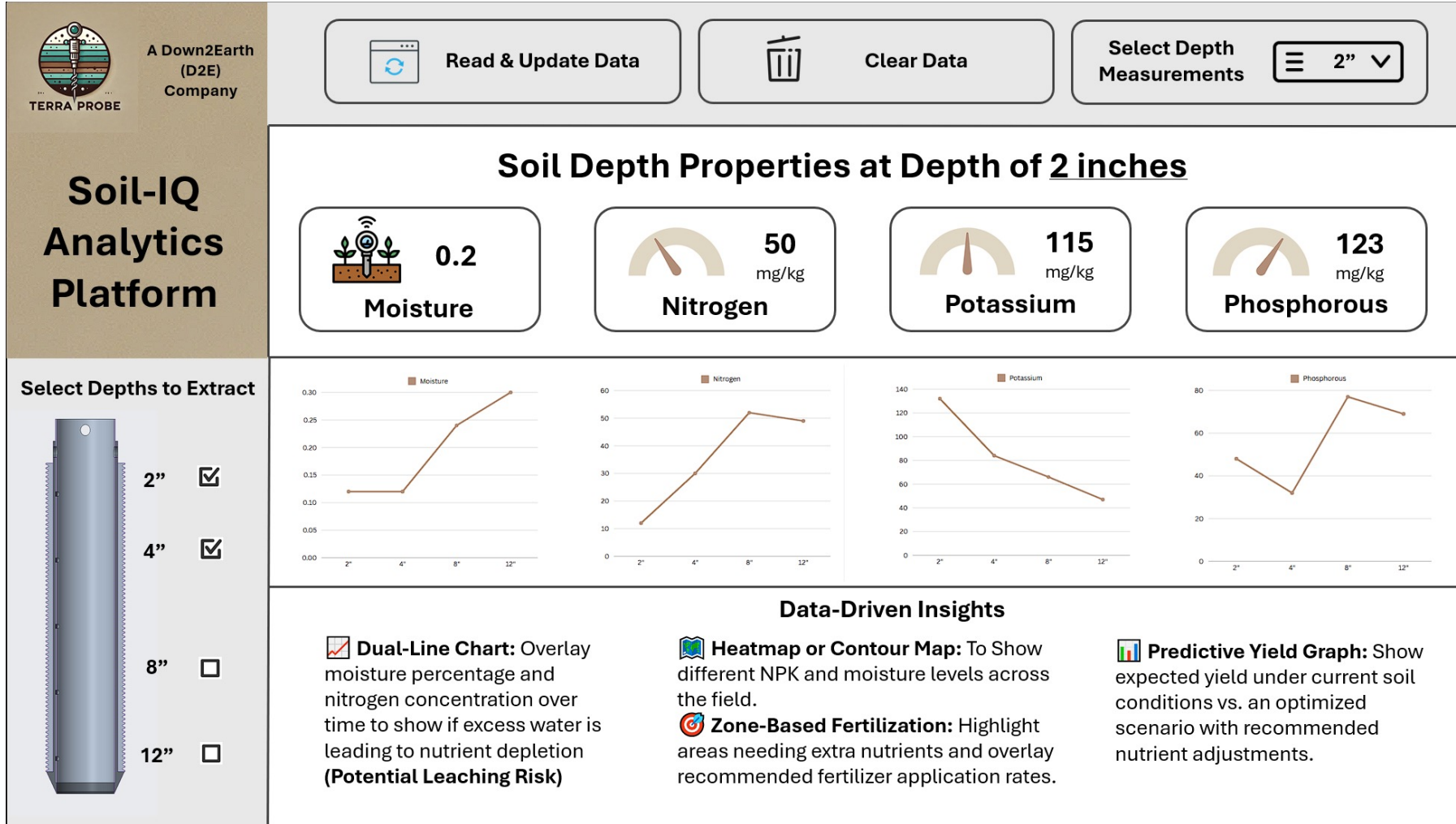
Subscription: \$12/month
per unit

Presenting an *Intermediate Solution*

- **Cost:** High-end solutions cost \$5,000+, while cheaper alternatives lack precision and automation.
- **Operation:** Instant soil collect and health insights, reducing analysis time from weeks to minutes
- **Portability:** Lightweight, automated, and no operator required
- **Recurring Revenue Model:** Sustainable model with consumables and data-driven services

TerraProbe Soil-IQ Dashboard

Digitally Enabling Competitive Advantage through Soil Health Analytics

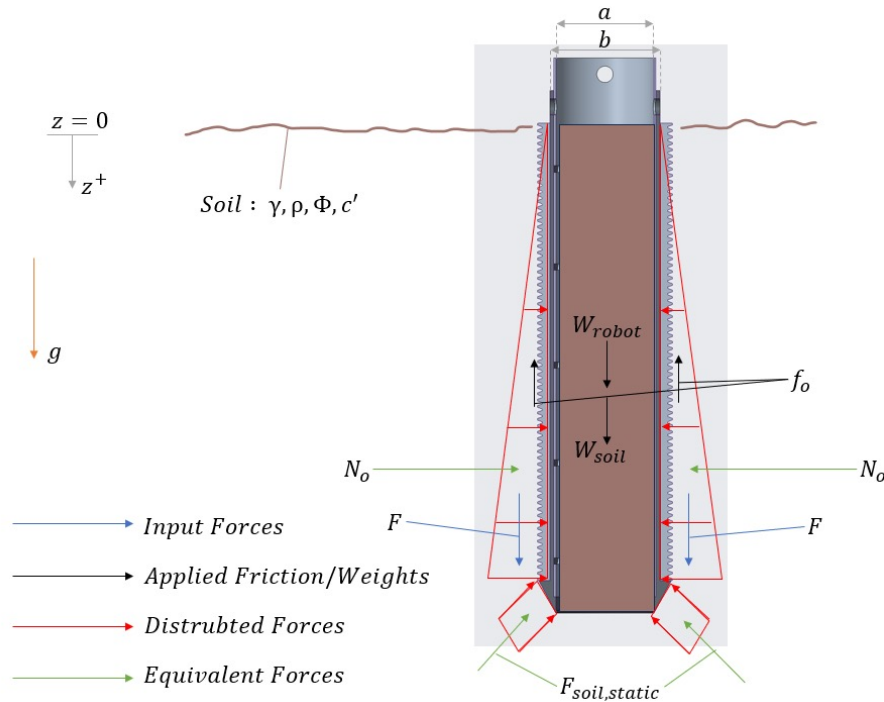


Soil-IQ Analytics

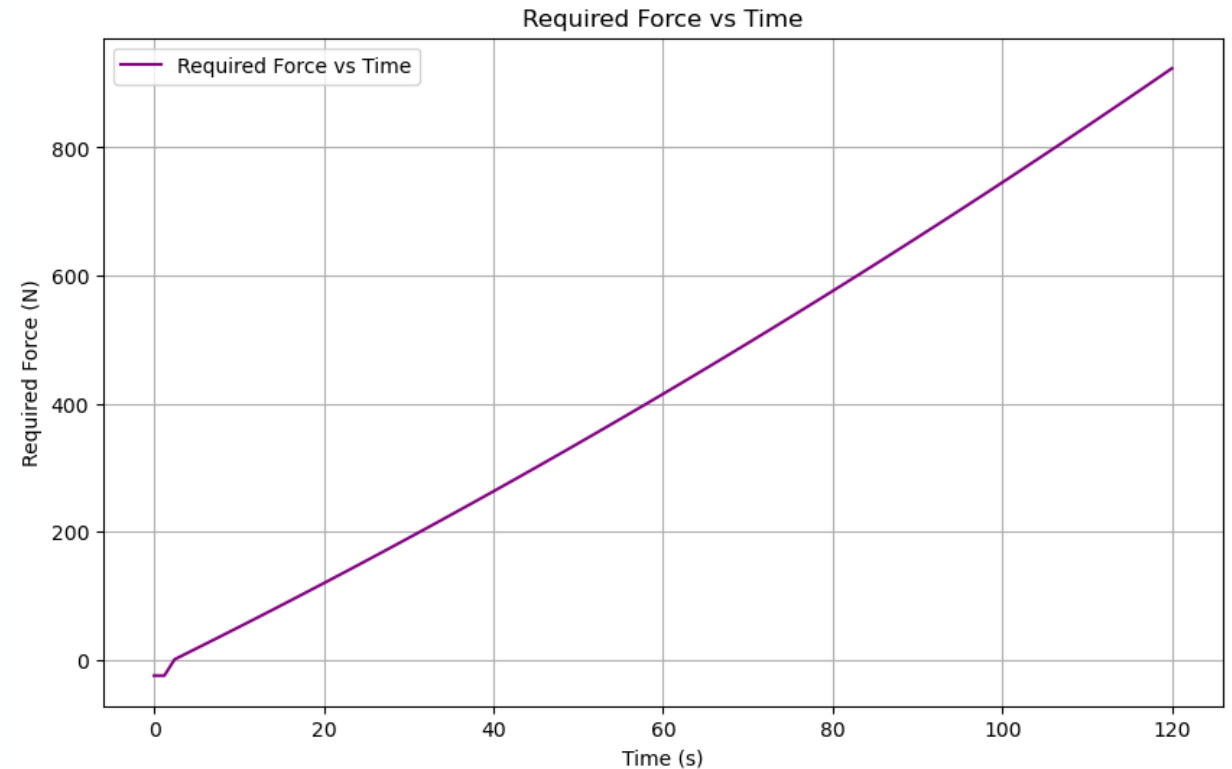
- **SD Card Integration:** Seamless transition from Testing Probe to Data Visualization
- **Real-Time Visualization:** Interactive dashboard powered by Python & Streamlit
- **Predictive Analytics:** Forecasts yield projections to optimize fertilizer application and crop yield.
- **User-Friendly Interface:** Simplified design enabling easy navigation and test measurements

Analysis: Pinion & Rack, Force Soil Modelling

The model was adjusted to the new requirements and the required force was determined



- Determined desired acceleration, velocity and position
- Calculated Force : **923 N**
- Motor requirements for the same : 35 W and 27.5 RPM

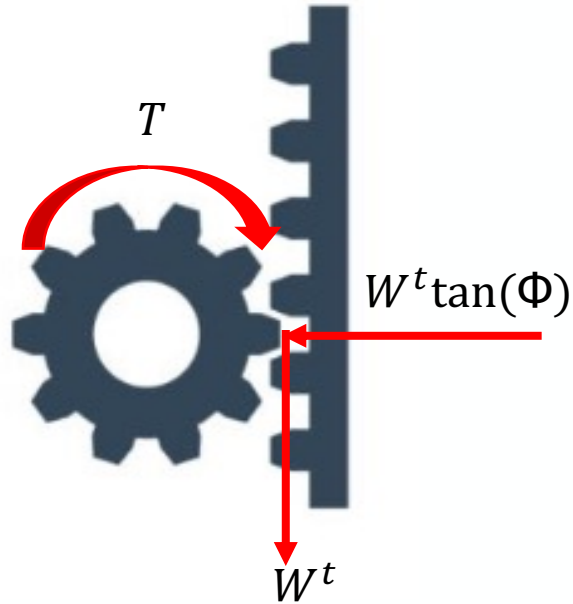


- The **Rankine Theory** : used to compute the lateral soil pressure on the sides of the guidance system
- Friction on the inside is considered negligible due to mechanical changes – more on this later

Analysis: Gear Failure

Steel 1144 Hardened can be used for the pinion – Factors of safety of 1.93 and 3.49 for wear and bending

Determining Loads:



$$W_T = 213.90 \text{ lbf} (958.29 \text{ N})$$

$$\Phi = 20^\circ, T = 213.90 \text{ lbf} - \text{in}$$

Determining Stresses:

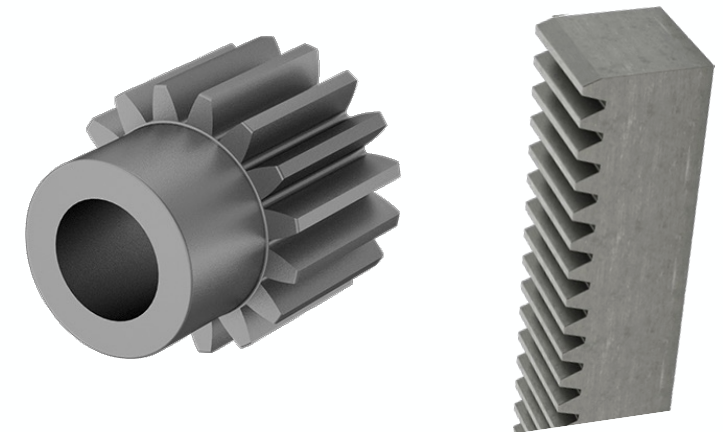
- **Wear:**
 $\sigma_c = 62.67 \text{ kpsi}$
- **Bending:**
 $\sigma_b = 26.56 \text{ kpsi}$

Determining Factors of Safety:

- **Wear:**
 $S_H = 1.93$
- **Bending:**
 $S_T = 3.49$

Pinion and Rack choices : *(All Choices follow AGMA Standards)

- Minimum # of Teeth – 16
- Pinion Diameter – 1 inch
- Rack/Pinion Pitch – 16 teeth/ inch
- Material : AISI 1144 Hardened Steel
- Parts from McMaster Carr



Analysis: Shaft Design

Shaft design optimized for strength, reliability, and seamless power transmission

Influence of Shaft, Key, & Bearings Design

Shaft (Diameter 0.5", Length 3" – AISI 4140 Steel)

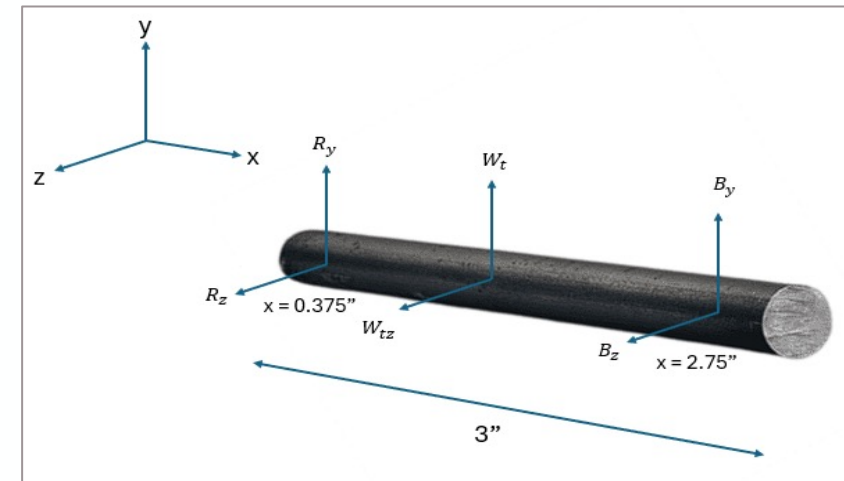
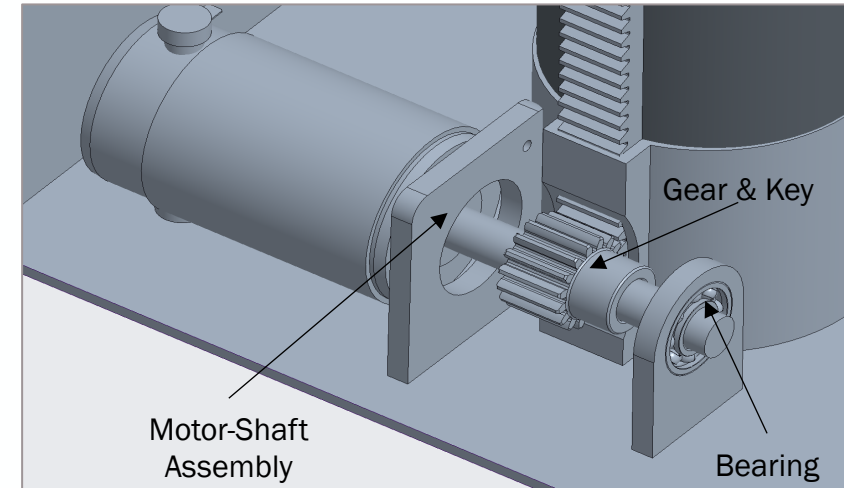
- Connects the motor to the gear system
- Fatigue Factor of Safety: 4.4
- Yielding Factor of Safety: 8.8

Key (0.125" x 0.125" x 0.25" – Square Key, AISI 1045 Steel)

- Critical for tight fit with the gear
- Shear & Crushing FOS: 1.5 (AGMA Standard Compliant)
- Torque Transmission: 106 lbf-in

Bearings (McMaster-Carr 60455K291)

- Prevents failures due to excessive stress, misalignment, or wear
- Static Load Capacity: 530 lbf
- Dynamic Load Capacity: 1140 lbf
- Max Speed: 25,000 RPM

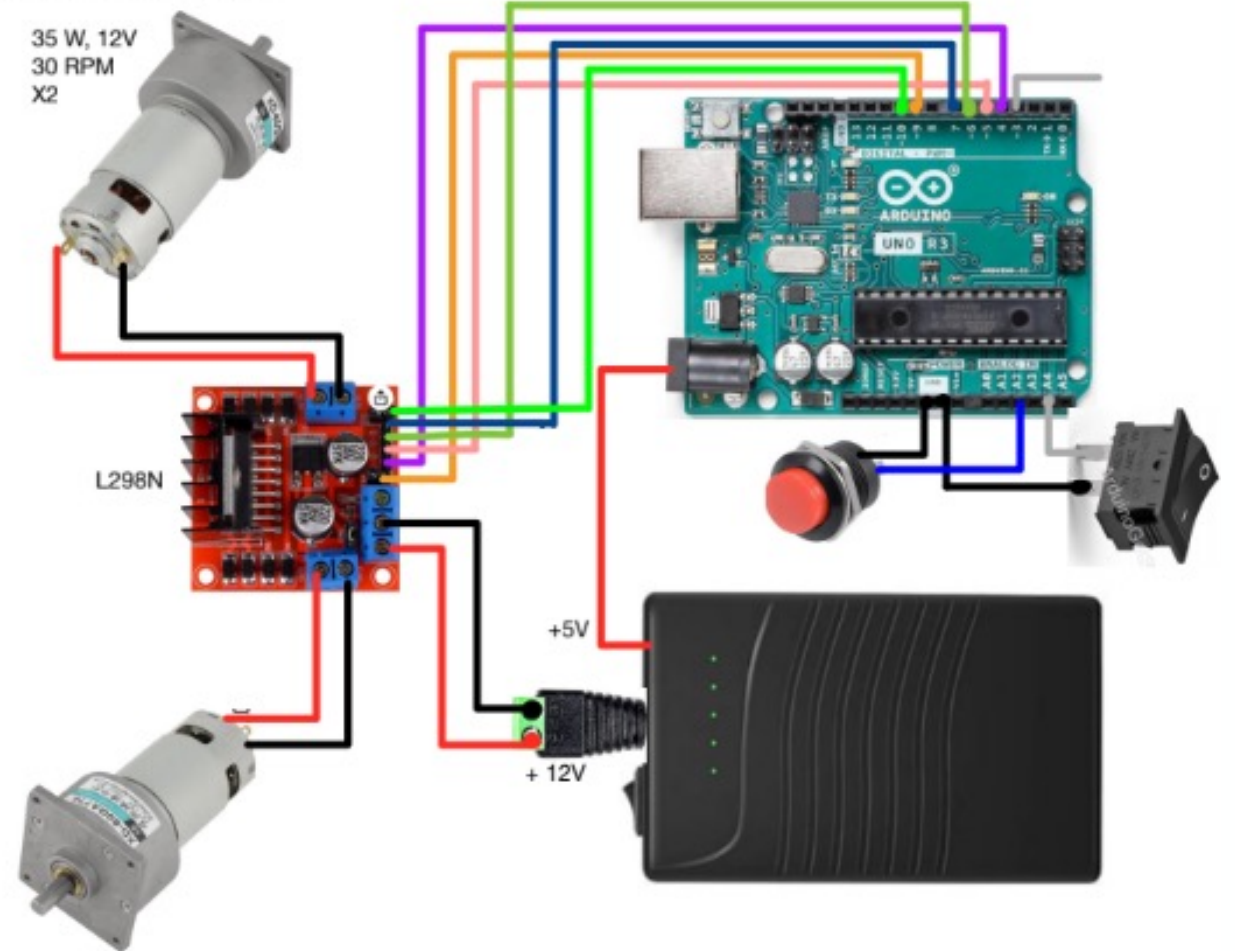


Motor Driver System - Electrical Diagrams

Controls power and direction for precise soil sampling operations

- Main Controller: Arduino Uno
 - 14 digital output pins
 - UART
- Motor Driver – L298N
- Module Control
 - Limit Switch Button
 - Emergency Stop Switch
- Motors
 - 35 W, 12 V
 - 30 RPM
 - High torque, Low RPM

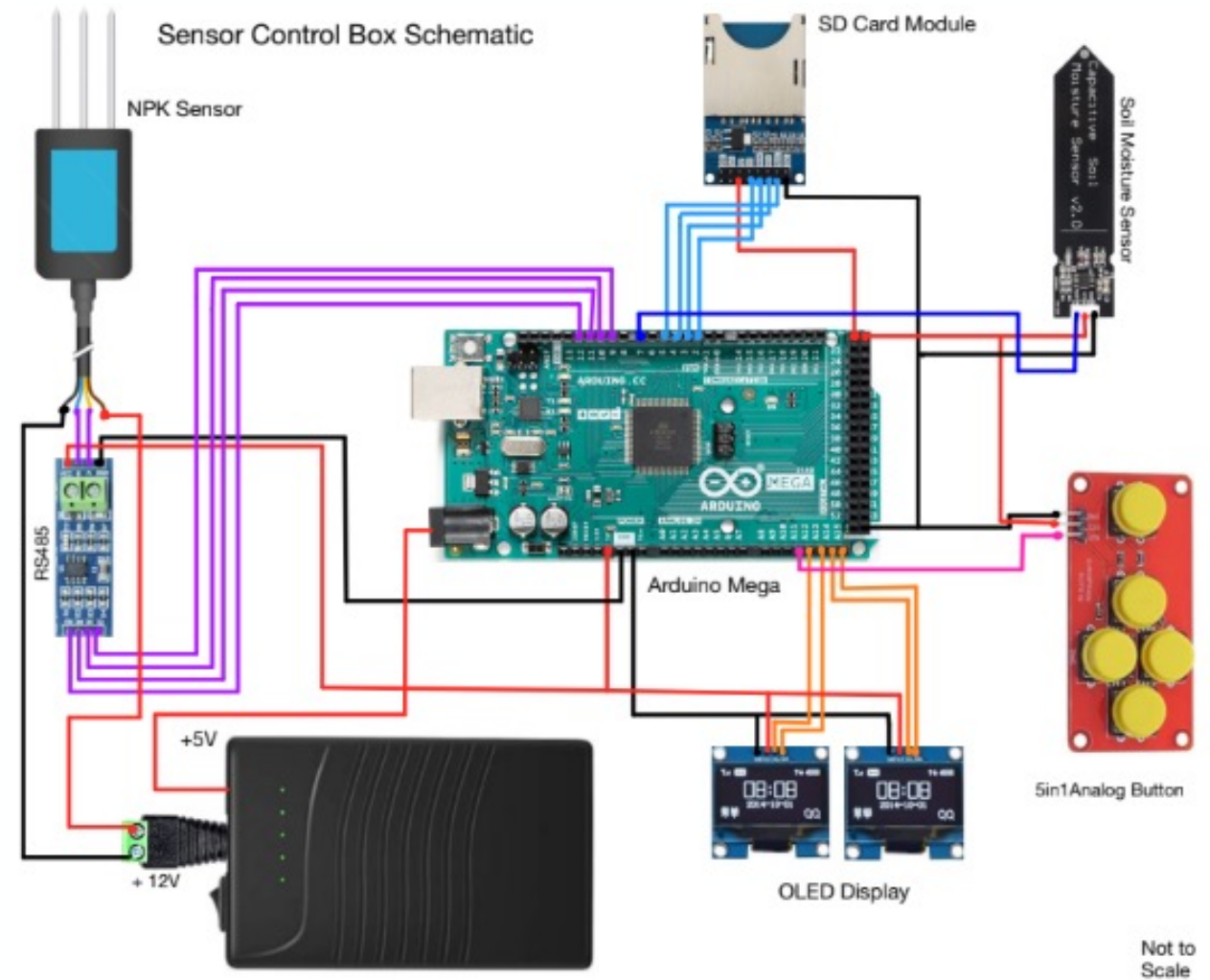
Motor Control Box Schematic



Data Acquisition System Electrical Diagrams

Captures real-time soil metrics for accurate analysis and decision-making

- Main Controller: Arduino Mega
 - 54 digital output pins
 - UART
- Data Collection
 - SD Card Module
 - 5in1 Analog Button
 - OLED Display – User Interface
- Sensors
 - Capacitive Soil Moisture
 - NPK (Nitrogen, Phosphorous, Potassium)
- Battery
 - 12 V – NPK
 - 5V – Arduino Uno



BOM: Materials & Costs / Sourcing Plan

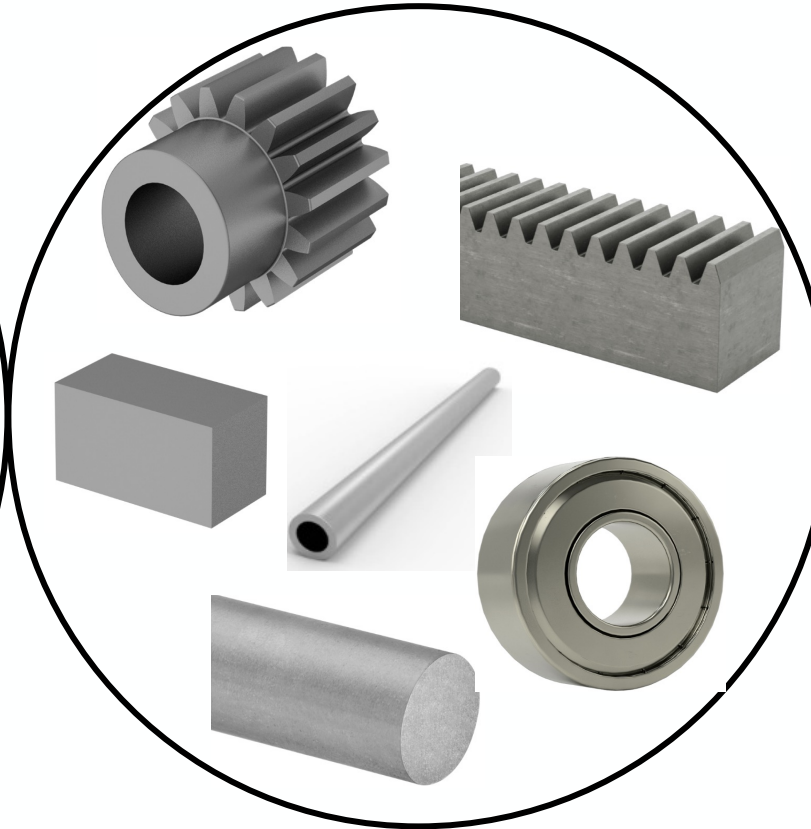
TerraProbe plans to spend ~\$448 to build its production prototype

Amazon



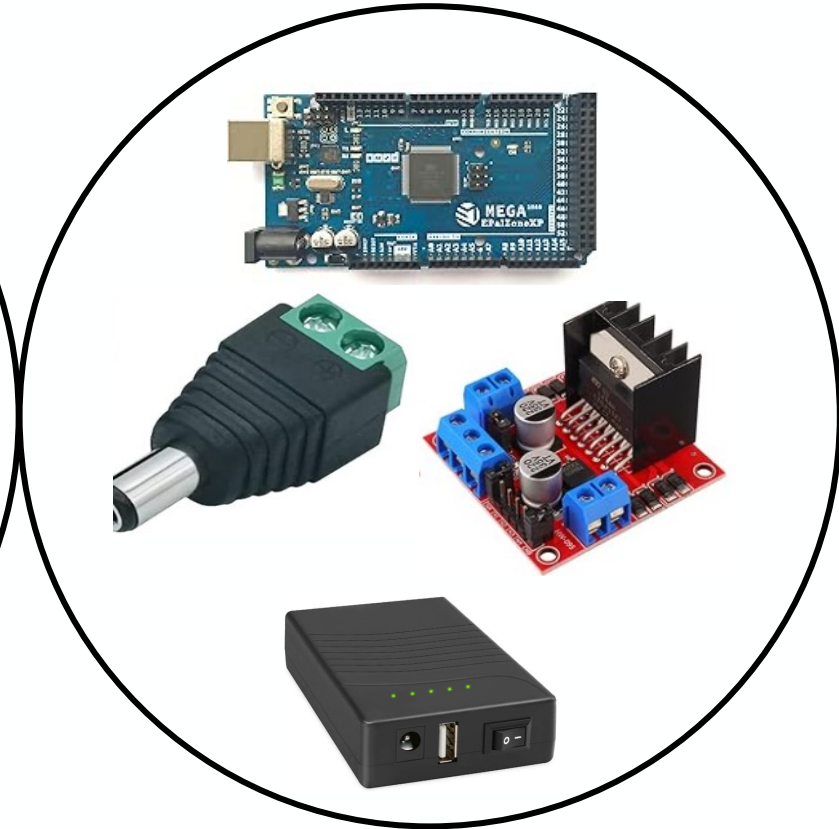
Subtotal: \$214.36

McMaster-Carr



Subtotal: \$233.83

Purdue ME Shop / Personal

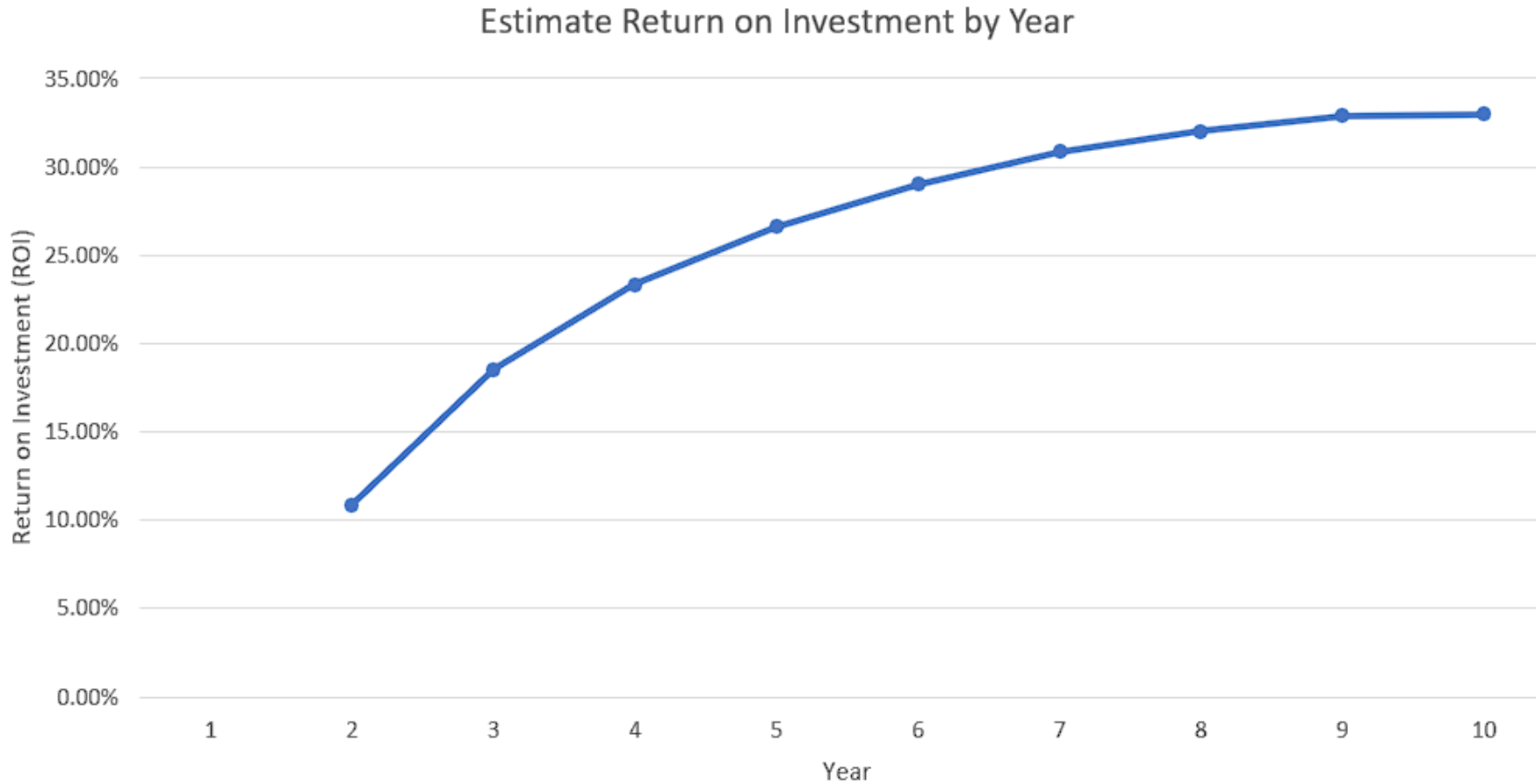


Subtotal: \$0.00

Total: \$448.19

Economic Analysis

TerraProbe presents strong scalability and projects to break even in ~3.5 years



Production Cost

TerraProbe → \$832.93

Inner Tube → \$84.24

Margin & Sell Price - TerraProbe

25% GM → \$1050 MRP

Margin & Sell Price – Inner Tube

50% GM → \$130 MRP

10-Year Annual Revenue

\$50M

10-Year ROI

33%

Manufacturing & Assembly Plans

Structured assembly process with in-house fabrication and precise sourcing to meet production goals

Component	Make/Buy	Justification
Inner Payload	Make	Requires specific shapes that are best controlled in-house.
Outer Payload	Make	Custom design requirements make in-house production preferable.
Sensor Probe Casing	Make	Custom casing ensures proper fit for sensors and electronics.
Foot Pedals	Make	Custom design is required to meet project needs.
Sensors	Buy	Precision components that are not feasible to manufacture.
Electronics	Buy	Specialized components requiring external sourcing.
Batteries	Buy	Standardized and regulated components best sourced externally.
Gears	Buy	Complex manufacturing process and high production cost.
Shafts	Buy	Bought at standard size and manufactured as needed by ME shop.
Motors	Buy	Precision-engineered components requiring external expertise.

Testing & Validation Plan

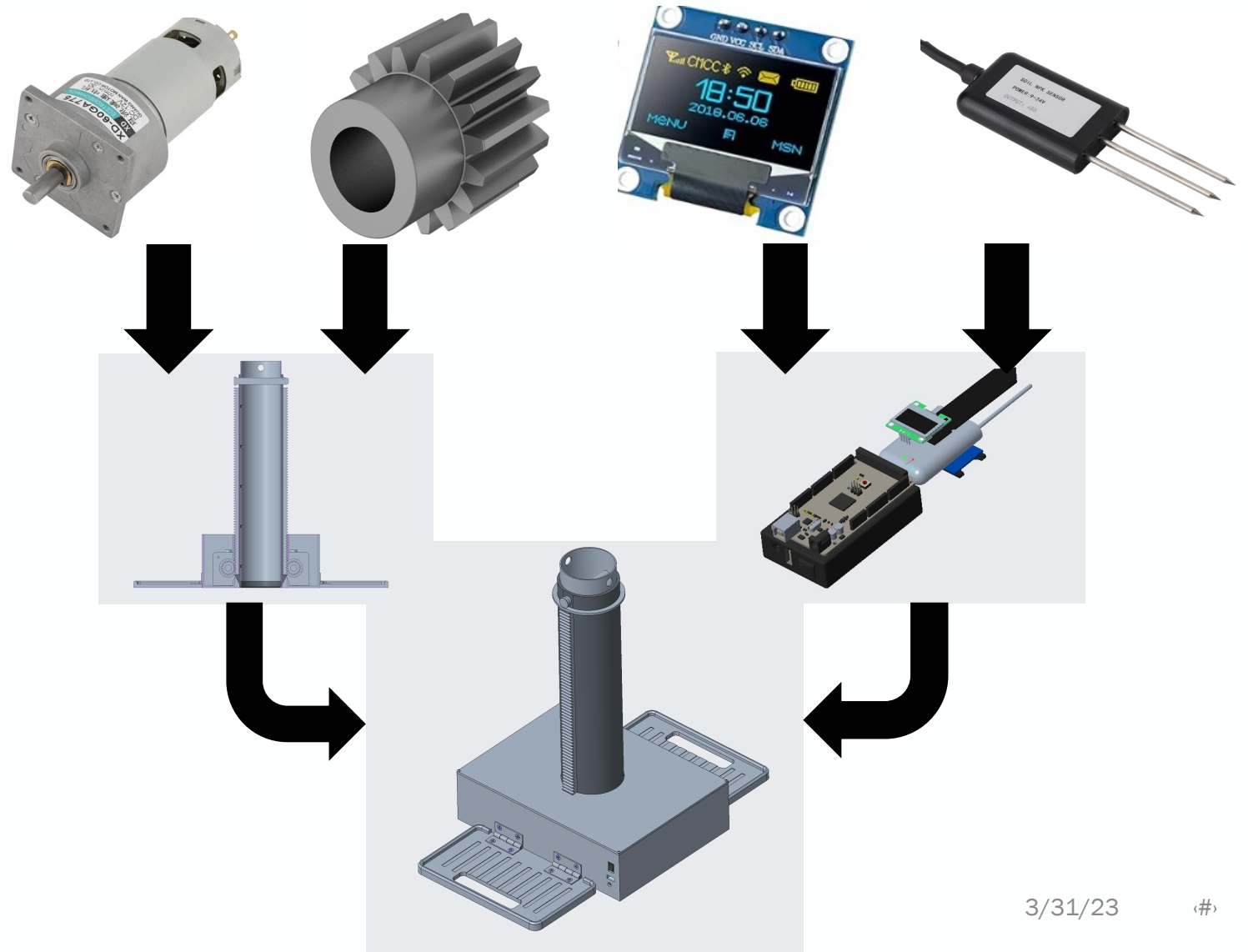
Testing individual components to subassemblies, culminating in full system integration

Preliminary Testing

- Motors
- Sensors
- OLED Display
- Key/Gear/Shaft

Subassembly Testing

- Motor-pinion-rack system
- Soil Collection
- Probe, data analytics



Current State & Path Forward

TerraProbe is on track for viable impact

Current State

- ☑ Project is **on schedule** – CAD & Mfg/Testing Plan Developed
- \$ ~\$215 spent (Low-Risk Parts) & \$785 of budget remains

Path Forward

- 🚚 Place Order for High-Risk Parts – Roughly \$200-350 in cost
- ☢ Testing (Sensor Accuracy/Calibration), Tolerances & Tight Fits

Business Impact

- 🏠 Intermediate– Low-end functionality and high-end features
- 🔥 Empower Small Farms – Data-driven decisions for better yields

Critical Design Review - Design		
PDR Prototype (Cardboard Build Testing Probe & Soil Burrowing)	2/2	2/6
Force Modelling for Soil Burrowing	2/6	2/10
Gear & Rack Analysis (Pitch Diameter, Teeth)	2/9	2/17
Shaft Design (Material, Size, Key, Bearings)	2/10	2/12
Failure Mode Effects Analysis (FMEA) & Risk Re-Evaluation	2/15	2/22
Manufacturing & Assembly Plan (Sub-Assemblies, Machining Processes)	2/22	2/28
Low-Risk Parts Order Placed	2/23	3/4
High-Level Testing Plan (Parts, Sub-Assemblies, Assembly)	3/2	3/5
Finalized Design & CAD Model	2/6	3/7
Final Design Review		
High-Risk Parts Ordered	3/5	3/15
Manufacturing Drawings & Operation Sheets	3/7	3/17
Discussion with Machine Shop and Testing of Sensors/Arduino/Motor	3/12	3/22
Sub-Assembly 1 (Shaft-Gear-Rack System) Manufacturing	3/16	3/31
Sub-Assembly 2 (Outside Casing, Foot Pedals) Manufacturing	3/31	4/7
Sub-Assembly (Soil Testing Probe) Manufacturing	4/7	4/12
Put Subassemblies Together (Test Device)	4/12	4/17
Testing, Marketing, Poster, & Quality Check	4/17	4/25
Individual Component Testing, Sub-Assembly & Assembly Testing	3/7	4/25

Thank You

Any Questions?



School of Mechanical Engineering

References

Following are used in the development of this CDR

Crop Nutrition. (n.d.). *Third-party soil sampling: What you need to know*. Retrieved

February 6, 2025, from <https://www.cropnutrition.com/resource-library/third-party-soil-sampling-what-you-need-to-know/>

North Carolina State University Extension. (n.d.). *Careful soil sampling: The key to reliable*

soil test information. Retrieved February 6, 2025, from <https://content.ces.ncsu.edu/careful-soil-sampling-the-key-to-reliable-soil-test-information>

Soil Science Society of America. (n.d.). *Self-study exam: Soil sampling and testing*.

Retrieved February 6, 2025, from <https://www.soils.org/files/certifications/certified/education/self-study/exam-pdfs/243.pdf>

California Department of Transportation. (2011). *Trenching and shoring manual*.

Retrieved February 6, 2025, from <https://engineering.purdue.edu/~frosch/ftp/Talbott/11%20-%20References/files/California%20Trenching%20and%20Shoring%20Manual.pdf>

CB Insights. (n.d.). *Agriculture tech trends*. Retrieved February 6, 2025,

from <https://www.cbinsights.com/research/agriculture-tech-trends/>

Engineering News-Record. (n.d.). *Market data*. Retrieved February 6, 2025,

from https://www.enr.com/market_data

Gartner. (n.d.). *Market research methodologies*. Retrieved February 6, 2025,

from <https://www.gartner.com/en/research/methodologies/market-research>

Grand View Research. (n.d.). *Soil testing market size, share & trends analysis report*.

Retrieved February 6, 2025, from <https://www.grandviewresearch.com/industry-analysis/soil-testing-market>

Grand View Research. (n.d.). *Smart agriculture market size, share & trends analysis report*.

Retrieved February 6, 2025, from <https://www.grandviewresearch.com/industry-analysis/smart-agriculture-farming-market>

References

Following are used in the development of this CDR

International Food Policy Research Institute. (n.d.). *Publications*. Retrieved February 6, 2025, from <https://www.ifpri.org/publications>

Mordor Intelligence. (n.d.). *Geotechnical engineering market - growth, trends, COVID-19 impact, and forecasts*. Retrieved February 6, 2025, from <https://www.mordorintelligence.com/industry-reports/geotechnical-engineering-market>

NASA. (n.d.). *Earth science*. Retrieved February 6, 2025, from <https://science.nasa.gov/earth-science/> United Nations Environment Programme. (n.d.). *Resources*. Retrieved February 6, 2025, from <https://www.unep.org/resources>

United States Department of Agriculture, Economic Research Service. (n.d.). *Home page*. Retrieved February 6, 2025, from <https://www.ers.usda.gov>

The World Bank. (n.d.). *Agriculture and rural development*. Retrieved February 6, 2025, from <https://data.worldbank.org/topic/agriculture-and-rural-development>

Envirotecnicos. (2020). *AMS soil sampling*. Retrieved February 6, 2025, from <https://www.envirotecnicos.com/wp-content/uploads/2020/07/AMS-soil-sampling-envirotecnicos.pdf>